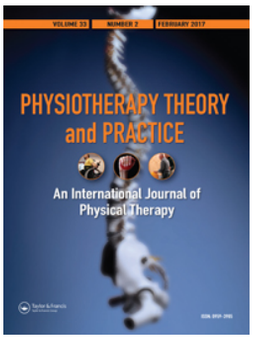

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RESEARCH REPORT

Electromyographic analysis of gluteus maximus and hamstring activity during the supine resisted hip extension exercise versus supine unilateral bridge to neutral

James W. Youdas, PT, MS[a], James P. Hartman, DPT[b], Brooke A. Murphy, DPT[c], Ashley M. Rundle, DPT[d], Jenna M. Ugorowski, DPT[a], and John H. Hollman, PT, PhD[a]

[a]Mayo Clinic School of Health Sciences, Rochester, MN, USA; [b]Winner Physical Therapy Inc., Winner, SD, USA; [c]HonorHealth Shea Medical Campus, Scottsdale, AZ, USA; [d]Fairview Health Services, Pine City, MN, USA

ABSTRACT

Hip extension strengthening exercises which maximize gluteus maximus contributions and minimize hamstring influences may be beneficial for persons with hip pain. This study's aim was to compare muscle activation of the gluteus maximus and hamstrings from healthy subjects during a supine resisted hip extension exercise versus supine unilateral bridge to neutral. Surface electromyographic (EMG) signals were obtained from the right gluteus maximus and hamstrings in 13 healthy male and 13 healthy female subjects. Maximum voluntary isometric contractions (MVICs) were collected to normalize data and permit meaningful comparisons across muscles. Peak median activation of the gluteus maximus was 33.8% MVIC for the bridge and 34.7% MVIC for the hip extension exercise, whereas peak median recruitment for hamstrings was 28.4% MVIC for the bridge and 51% MVIC for the hip extension exercise. The gluteus maximus to hamstrings ratio was compared between the two exercises using the Wilcoxon signed-ranks test ($\alpha = 0.05$). The ratio ($p = 0.014$) was greater in the supine unilateral bridge (median = 111.3%) than supine hip extension exercise (median = 59.2%), suggesting a reduction of hamstring recruitment in the unilateral bridge to neutral compared to the supine resisted hip extension exercise. The supine hip extension exercise demonstrated higher EMG activity of hamstrings in comparison with supine unilateral bridge and, therefore, may be less appropriate in subjects who need to increase gluteus maximus activation.

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KEYWORDS
Electromyography; gluteus maximus; hamstrings

Introduction

The principal extensors of the hip joint are the gluteus maximus, hamstrings (long head of biceps femoris, semimembranosus, and semitendinosus), and posterior head of the adductor magnus (Neumann, 2010). To develop an effective strengthening program for the hip extensors, a rehabilitation specialist should consider several biomechanical factors including: 1) external moment arm length; 2) gravitational effects; 3) base of support; and 4) patient position (Reiman, Bolgla, and Loudon, 2012). Previously, investigators reported electromyographic (EMG) analyses of hip extensors in healthy persons using surface electrodes for a variety of rehabilitation exercises (Ayotte, Stetts, Keenan, and Greenway, 2007; Ekstrom, Donatelli, and Carp, 2007; Ekstrom, Osborn, and Hauer, 2008). Common subject positions for exercising hip extensors were supine (i.e., bridging) and standing (i.e., lunges, step-ups, and squats) (Ayotte, Stetts, Keenan, and Greenway, 2007; Ekstrom, Donatelli, and Carp, 2007; Ekstrom, Osborn, and Hauer, 2008). Gluteus maximus (Table 1) and hamstring (Table 2) muscle recruitment levels for an assortment of accepted rehabilitation exercises were normalized to each muscle's maximum voluntary isometric contraction (MVIC) and expressed as a percentage (% MVIC). Gluteus maximus EMG activity ranged from 88% (SD 24) to 20% (SD 10) MVIC, whereas hamstring recruitment ranged from 82% (SD 24) to 9% (SD 4) % MVIC (Ayotte, Stetts, Keenan, and Greenway, 2007; Ekstrom, Osborn, and Hauer, 2008).

In a series of modeling studies, investigators (Lewis, Sahrmann, and Moran, 2007; Lewis, Sahrmann, and Moran, 2009) reported decreased force production from the gluteal muscles during hip extension yielded increased net resultant forces in the anterior direction at the hip joint. This in turn may be a potential cause of anterior hip joint pain or disability (Shindle, Ranawat, and Kelly, 2006). One supine exercise not previously studied using surface EMG, yet described in a popular muscle testing textbook (Hislop, Avers, and Brown, 2014) is the supine resisted hip

CONTACT James W. Youdas youdas.james@mayo.edu Mayo Clinic College of Medicine, 200 1st St. SW, Rochester, MN 55905, USA
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Table 1. Gluteus maximus percent maximum voluntary isometric contraction (% MVIC) ranking of exercises.

	Exercise	Gluteus maximus % MVIC
1.	Maximum resistance to prone trunk extension to end range with lower extremities stabilized (Ekstrom et al, 2008)	88 ± 15
2.	Wall squat (Ayotte et al., 2007)	86 ± 43
3.	Forward step-up (Ayotte et al., 2007)	74 ± 43
4.	Opposite upper and lower extremity lifts with neutral spine in the quadruped position with maximal manual resistance to the lifted lower extremity at the distal leg (Ekstrom et al., 2008)	68 ± 23
5.	Retro step-up (Ayotte et al., 2007)	59 ± 35
6.	Mini-squat and lateral step-up (Ayotte et al., 2007)	57 ± 29
7.	Quadruped arm/lower extremity lift (Ekstrom et al., 2007)	56 ± 22
8.	Unilateral bridge (Ekstrom et al., 2007)	40 ± 20
9.	Lunge (Ekstrom et al., 2007)	36 ± 17
10.	Lateral step-up (Ekstrom et al., 2007)	29 ± 13
11.	Double leg bridge (Ekstrom et al., 2007)	25 ± 14
12.	Active hip abduction (Ekstrom et al., 2007)	21 ± 16
13.	Bridge to neutral spine position with knees extended and feet on gymnastics ball (Ekstrom et al., 2008)	20 ± 10

Table 2. Hamstring percent maximum voluntary isometric contraction (% MVIC) ranking of exercises.

	Exercise	Hamstrings % MVIC
1.	Maximum resistance to prone trunk extension to end range with lower extremities stabilized (Ekstrom et al., 2008)	82 ± 24
2.	Opposite upper and lower extremity lifts with neutral spine in the quadruped position with maximum manual resistance to the lifted lower extremity at the distal leg (Ekstrom et al., 2008)	67 ± 16
3.	Unilateral bridge (Ekstrom et al., 2007)	40 ± 17
4.	Quadruped arm/lower extremity lift (Ekstrom et al., 2007)	5± 14
5.	• Bridge to neutral spine position with knees extended and feet on gymnastics ball (Ekstrom et al., 2008)	35 ± 14
	• Bridge to neutral spine (Ekstrom et al., 2008)	35±21
6.	Double leg bridge to neutral (Ekstrom et al., 2007)	24 ± 14
7.	Wall squat (Ayotte et al., 2007)	15 ± 8
8.	Forward step-up (Ayotte et al., 2007)	9± 6
9.	• Lunge (Ekstrom et al., 2007)	11 ± 6
	• Mini-squat (Ayotte et al., 2007)	11 ± 6
10.	Retro step-up (Ayotte et al., 2007	10 ± 5
11.	Lateral step-up (Ayotte et al., 2007)	9 ± 4

extension exercise (Figure 1). This procedure requires simultaneous recruitment of the gluteus maximus and hamstrings when manual resistance is applied to the subject's heel by an examiner in the direction of hip flexion. The opposite lower extremity will automatically elevate demonstrating a braced pelvis (Hislop, Avers, and Brown, 2014).

In patients with anterior hip joint pain, the aim of rehabilitation is to minimize or eliminate pain (Lewis, Sahrmann, and Moran, 2007; Lewis, Sahrmann, and Moran, 2009). This could be accomplished by increasing the gluteus maximus (% MVIC) to hamstring (% MVIC) ratio. One exercise thought to enhance the gluteus maximus to hamstring ratio is the supine unilateral bridge to neutral (Neumann, 2010; Worrell, Karst, Adamczyk, and Moore, 2001). Clinicians suggest that the supine unilateral bridge exercise may reduce recruitment of the hamstrings because the knee joint of the weight-bearing lower extremity is flexed to about 90° which reduces hamstring muscle length and potential hip extension force production.

The purpose of this study was to compare the gluteus maximus (% MVIC) to hamstrings (% MVIC) ratio between supine unilateral bridge to neutral spine and the supine hip extension exercise. We hypothesized that the gluteus maximus to hamstrings ratio would be greater during supine unilateral bridge compared to the supine hip extension exercise because of reduced hamstring muscle activation in the bridge since the knee joint was flexed to 90°. The basis of our hypothesis was the presence of knee extension during the supine hip extension exercise which would favor recruitment of the hamstrings.

Methods

Subjects

Twenty-six healthy subjects volunteered to participate. Each subject self-reported the number of days during the week when he or she participated in a personal workout session consisting of aerobic exercise, strength training, or both. Subject demographics and anthropometrics are summarized in Table 3. Subjects comprised a sample of convenience and were recruited from the Mayo Clinic School of Health Sciences. To be eligible to participate in this study, volunteers needed to be between 20 and 39 years of age. Volunteers reporting a history of the following lower extremity conditions were excluded from the study: 1) previous subluxation, dislocation, or fracture; 2) joint instability, tendinitis, bursitis, impingement, adhesive capsulitis, neurovascular complications, or any condition that limited physical activity for greater than two days over the last six months; and 3) current complaints of neuromuscular pain, numbness, or tingling in the lower extremity and back. Persons with acute low back pain were excluded from participating in the study because back pain could prevent correct performance of the supine exercises. A sample size of 22 subjects was required to detect a mean difference in EMG recruitment of 10% MVIC (effect size = 0.20) between conditions with a statistical power (1-ß) equal to 0.80 at a = 0.05 (Faul, Erdfelder, Lang, and Buchner, 2007). Written informed consent was obtained prior to the start of data collection, and study procedures were approved by the institutional review board.

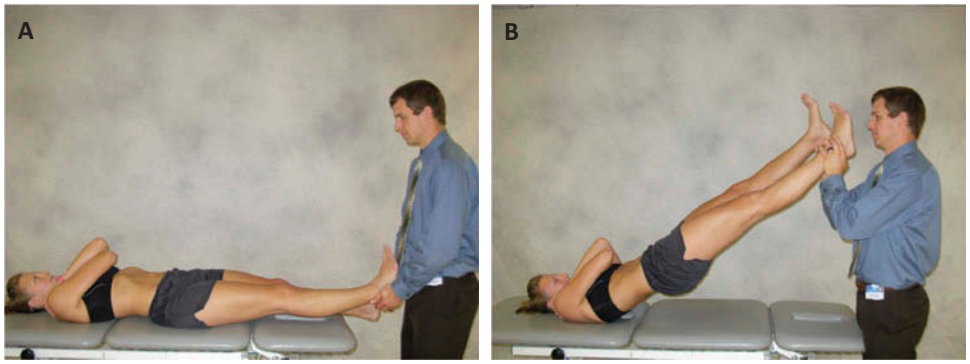

Figure 1. Supine resisted hip extension exercise for the right lower extremity. (A) The start position occurs with the subject supine on a table with arms folded across the chest. The examiner stands at the foot of the table with both hands cupped under the subject's right heel. The subject presses his/her heel into the examiner's cupped hands. (B) The examiner instructs the subject to not permit him to elevate the right lower extremity from the surface of the table and to keep the hip from moving. The examiner lifts the right lower extremity about 90 cm [35 inch] off the table's surface to the level of his sternoclavicular joint. The subject's pelvis and back are elevated as a stiffened unit, while the left lower extremity parallels the right lower extremity.

Table 3. Demographics and anthropometrics of subjects.

	Age (y)			Height (m)			Mass (kg)			Body mass index (kg/m^2)			Days/week of physical activity		
Gender	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range
Male (n=13)	23.4	1.3	22 – 26	1.8	0.1	1.7 – 2	79.7	10.6	58 – 97	24	3	19.2 – 30.6	4.9	1.2	3 – 7
Female (n=13)	23.5	1.2	22 – 27	1.7	0.1	1.6 – 1.8	63.7	7.4	56 – 83	22	2.2	19.3 – 26.2	5.2	0.8	4 – 6

Instrumentation

Raw EMG signals were collected with Bagnoli™ DE 3.1 double-differential surface EMG sensors (Delsys Inc., Boston, MA, USA). The sensor contacts were made from 99.9% pure silver bars 10 mm in length, spaced 10 mm apart, and encased within preamplifier assemblies measuring 41 × 20 × 5 mm. The preamplifier gain was 10 v/v. The combined preamplifier and main amplifier permitted a gain from 100 to 10,000 Hz. The common-mode rejection ratio was 92 dB at 60 Hz, and input impedance was greater than 1,015 Ω at 100 Hz. Data were sampled at a frequency of 1,000 Hz. Raw EMG signals were processed with EMG works® Data Acquisition and Analysis software (Delsys Inc.).

Testing procedure

Data were collected in a research laboratory. All subjects wore shorts, tee shirts, or sports bras to permit correct placement of the EMG electrodes. Electrode placement for the gluteus maximus was over the center of the muscle belly between the lateral edge of the sacrum and posterior edge of the greater trochanter (Criswell, 2011). A general placement for the hamstring group was used with the electrode situated midway between the gluteal fold and the popliteal fossa (Criswell, 2011).

The skin over each muscle belly was prepared by shaving hair from the vicinity and cleansing with isopropyl alcohol wipes. Surface electrodes were attached to the cleansed area with adhesive interfaces (Delsys Inc.) and secured with 3M Transpore™ medical tape (St. Paul, MN, USA). A common ground electrode was placed over the medial malleolus of the left lower extremity. Wires from the electrodes were connected to a small input module attached to a gait belt secured around the patient's waist. Next, subjects performed resistance tests using five-second hold times to provide subjects several practice trials so they were familiar with the muscle test procedures. Specific muscle test procedures have been previously described (Hislop, Avers, and Brown, 2014).

Maximum voluntary isometric contractions (MVICs) for each muscle test procedure were collected. For each subject, examiners provided verbal encouragement to help the subject produce maximum effort during the MVIC. The gluteus maximus was tested in prone. The examiner instructed the subject to extend the right thigh with the knee flexed to 90 degrees by raising the sole of the foot toward the ceiling. The examiner applied maximum manual resistance just proximal to the popliteal fossa in an inferior direction creating a hip flexion external moment (Figure 2A). The knee joint angle was controlled visually by an examiner.

The hamstrings were tested with the subject prone and right hip in neutral rotation and leg flexed to 45

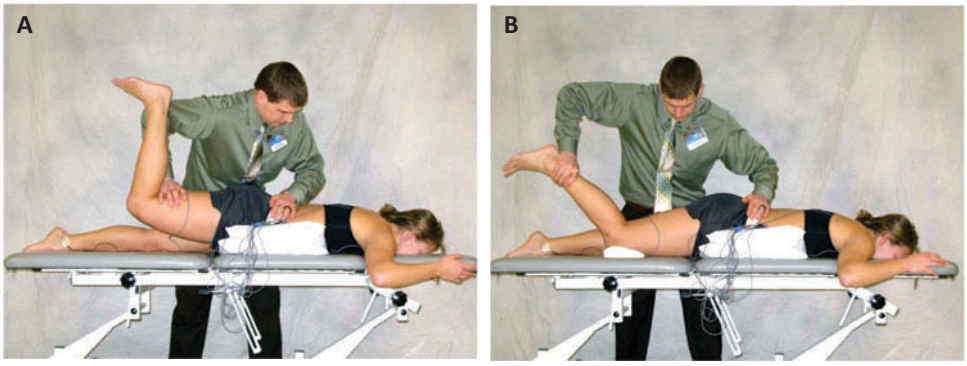

Figure 2. Manual muscle test positions. (A) Muscle test position for the gluteus maximus. (B) the hamstrings in aggregate.

degrees. The examiner placed a rolled towel just proximal to the patella to avoid excessive contact with the support surface. Next, the examiner grasped the subject's right calcaneus and attempted to extend the leg, while the subject maintained the knee joint at 45 degrees of knee flexion (Figure 2B). The knee joint angle was controlled visually by an examiner.

Prior to testing, each participant was provided a visual demonstration of the exercises by an examiner followed by a series of practice repetitions to ensure that the subject demonstrated correct body mechanics during performance of each exercise. During the testing session, subjects performed three repetitions of the exercise. A metronome was set at 40 beats per minute to standardize the rate of movement across subjects. A one-minute rest between repetitions was used to avoid fatigue.

Supine hip extension exercise

The supine hip extension exercise began with the subject supine on a table with arms folded across the chest (Hislop, Avers, and Brown, 2014). The examiner stood at the foot of the table with both his hands cupped under the right heel of the subject's test lower extremity. The right limb's knee was maintained in extension and hip in neutral (Figure 1A). Next, the subject was instructed to press his/her right heel into the examiner's cupped hands. The subject was trained to resist the examiner from lifting the right lower extremity from the surface of the table and to keep the right hip from moving. The examiner lifted the subject's right lower extremity about 90 cm vertically [35 inch] from the table's surface (Figure 1B), so the heel was placed at the level of the examiner's sternoclavicular joint. The elevated position was maintained for five seconds. The subject's pelvis and back elevated as a stiffened unit, while the left lower extremity paralleled the right lower extremity which revealed that the pelvis was maintained in a neutral position by isometric activation of trunk and pelvic muscles.

Supine unilateral bridge to neutral spine

The supine unilateral bridge to neutral spine exercise was performed using the subject's right lower extremity (Figure 3A). The participant's hands were placed in contact with the floor with palms facing down. Upon the examiner's command, the subject pressed the right foot against the support surface and elevated the buttocks from the floor to attain a straight line between the medial–lateral axes of the shoulder, hip, and knee joints. The knee joint angle was estimated visually by an examiner to be at 90° at the end point of the unilateral bridge (Figure 3B). The left hip remained in neutral and left knee remained extended (Youdas et al., 2015). This exercise position was maintained for 10 seconds.

Data reduction

The dependent variable was normalized peak EMG activity (% MVIC) for the gluteus maximus and hamstrings. Raw EMG data collected during the tests were band-pass filtered between 20 and 450 Hz and subsequently processed with a root-mean-square algorithm using moving windows with 125 ms time constants. EMG data collected during the supine resisted hip extension test and supine unilateral bridge were normalized to each muscle's respective MVIC trials and therefore expressed as a percentage of the MVIC (% MVIC). Peak activation for each muscle was calculated from the normalized data using a 200-ms window about the peak.

To assist with classification of muscle recruitment from very high to low, we used a scheme described in 1992 (DiGiovine, Jobe, Pink, and Perry, 1992). The categorization is as follows: very high is >60% MVIC;

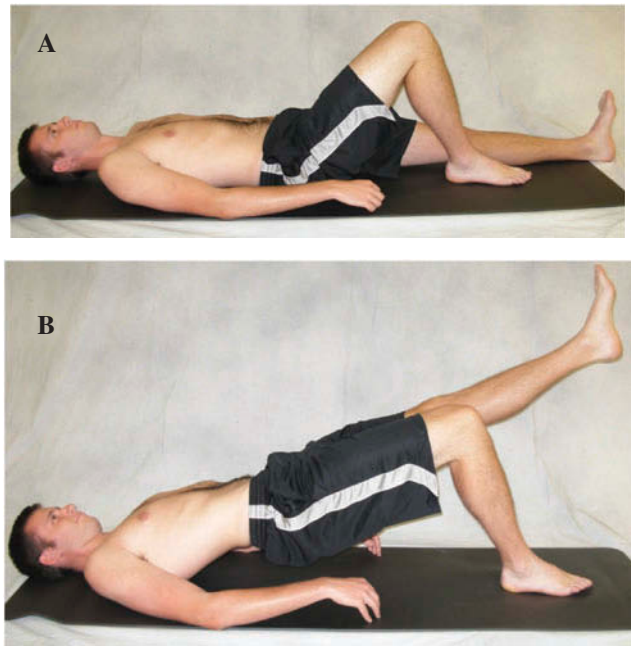

Figure 3. Single leg bridging to neutral exercises. The hands are placed on the floor by the sides with palms facing down for all exercise positions. (A) The start position for single leg bridge to neutral begins with the right lower extremity in a hook-lying position and the plantar surface of the foot flat on the support surface. The hip of the left lower extremity is in neutral with the knee in terminal extension and the foot unsupported. (B) The end position of a single leg bridge occurs with the pelvis elevated from the support surface with the right hip and spine in neutral and knee flexed approximately 90° with the plantar surface of the foot flat on the support surface. The left lower extremity remains non-weight-bearing.

high is 41–60% MVIC; moderate is 21–40% MVIC; and low is 0–20% MVIC.

Statistical analysis

Descriptive statistics including distribution shapes, measures of central tendency, and variability of the EMG recruitment data were calculated. Three of four distributions were positively skewed (Shapiro–Wilk tests were statistically significant at $p < 0.05$), so data were analyzed with nonparametric statistics. Median values and interquartile ranges were used to characterize central tendency and variability. Magnitudes of EMG recruitment in the gluteus maximus and hamstrings were compared with the Wilcoxon signed-ranks test ($\alpha = 0.05$). For each participant, we also divided gluteus maximus % MVIC by the hamstrings % MVIC to compute the ratio (%) of peak activation of gluteus maximus to hamstrings for both the supine hip extension exercise and the supine unilateral bridge to neutral spine. Data were analyzed with IBM® SPSS® Statistics 21.0 software (IBM Corp, Armonk, NY).

Table 4. Descriptive statistics for median peak values of muscle recruitment of gluteus maximus and hamstrings during supine hip extension exercise and the supine unilateral bridge to neutral spine.

Variable	Percentiles 25th	50th median	75th	Wilcoxon signed-ranks test (median)
Supine Hip Extension – GMAX (% MVIC)	17.9	34.7	49.4	p = 576
Unilateral Bridge – GMAX (% MVIC)	26.9	33.8	42.7	
Supine Hip extension – HS (% MVIC)	32.8	51	77.6	p = 004
Unilateral Bridge – HS (% MVIC)	18.3	28.4	46.4	
Supine Hip Extension – GMAX to HS ratio (%)	36.7	59.2	123.3	p = 014
Unilateral Bridge – GMAX to HS ratio (%)	78.3	111.4	215.1	

MVIC – maximum voluntary isometric contraction
GMAX – gluteus maximus
HS – hamstrings
Bridge – unilateral supine bridge to neutral
Hip extension – supine hip extension exercise

Results

Descriptive information is displayed (Table 4) for the median peak value and interquartile ranges of muscle recruitment (% MVIC) for gluteus maximus and hamstrings during performance of the supine resisted hip extension exercise and supine unilateral bridge to neutral spine. Gluteus maximus recruitment did not differ between exercises ($p = 0.576$). Hamstring activation was reduced in the unilateral bridge compared to the supine hip extension exercise ($p = 0.004$). Gluteus maximus to hamstring ratio was greater ($p = 0.014$) in the unilateral bridge than in the supine hip extension exercise.

Discussion

The purpose of this study was to inform rehabilitation professionals of the gluteus maximus (% MVIC) to hamstrings (% MVIC) ratio between supine unilateral bridge to neutral and the supine hip extension exercise. The gluteus maximus to hamstrings ratio was significantly greater during supine unilateral bridge to neutral spine than the supine hip extension exercise because of reduced hamstring muscle activation in the unilateral bridge exercise. This finding supports our research hypothesis. The supine hip extension exercise would not be an appropriate exercise for subjects with anterior hip pain because of the reduced gluteus maximus to hamstrings ratio.

Currently, the literature does not describe the gluteus maximus to hamstring muscle activation ratio for a variety of exercises prescribed by rehabilitation specialists for persons with anterior hip joint pain. Nevertheless, the literature does supply muscle activation values (% MVIC) of the gluteus maximus (Table 1) and hamstrings (Table 2) for eight common weight-bearing exercises. We used these data to calculate the gluteus maximus to hamstring ratios (Table 5). Five unilateral weight-bearing exercises demonstrated gluteus maximus to hamstring ratios ranging from 622% to 518% (Ayotte, Stetts, Keenan, and Greenway, 2007). Gluteus maximus recruitment ranged from high (41–60% MVIC) to very high (>60% MVIC), whereas all hamstring muscle activation levels were low (<20% MVIC). These five exercises illustrate selected muscle recruitment of the gluteus maximus with diminished activation of the hamstrings. Another group of investigators studied both double and single leg bridge to neutral spine as well as a standing lunge (Ekstrom, Donatelli, and Carp, 2007). Gluteus maximus to hamstring ratios varied from 327% to 100% with the lunge ratio three times greater than either double or single leg bridge. According to data from Ekstrom Donatelli, and Carp (2007), the gluteus maximus and hamstrings were recruited equivalently for the unilateral bridge to neutral spine. This finding compares favorably with data from the present study since we reported a gluteus maximus to hamstring ratio of 111.4% for the supine unilateral bridge to neutral. Furthermore, in contrast to data reported by Ayotte, Stetts, Keenan, and Greenway (2007), neither exercise examined in the present study required high to very high levels of gluteus maximus muscle recruitment.

Table 5. Recruitment [% maximum voluntary isometric contraction (MVIC)] of the gluteus maximus and hamstrings during common lower extremity exercises.

Exercise	Gluteus maximus % MVIC	Hamstrings % MVIC	Ratio of gluteus maximus to hamstrings (%)
*Unilateral lateral step-up (Ayotte et al., 2007)	56 ± 29	9 ± 4	622
*Unilateral forward step-up (Ayotte et al., 2007)	74 ± 43	12 ± 6	616
*Unilateral retro step-up (Ayotte et al., 2007)	59 ± 35	10 ± 5	590
*Unilateral wall squat (Ayotte et al., 2007)	86 ± 43	15 ± 8	573
*Unilateral mini-squat (Ayotte et al., 2007)	57 ± 29	11 ± 6	518
Lunge (Ekstrom et al., 2007)	36 ± 17	11 ± 6	327
Double leg bridge to neutral spine (Ekstrom, 2007)	25 ± 14	24 ± 14	104
*Unilateral bridge to neutral spine (Ekstrom, 2007)	40 ± 20	40 ± 17	100

*Indicates exercise involved unilateral weight-bearing.

Study limitations

The primary limitation of this study is our use of healthy participants. Additional research should be performed on patients with anterior hip joint pain during exercise. We do not know whether subjects with anterior hip joint pain could perform the unilateral supine unilateral bridge to neutral spine pain-free, while the supine hip extension test elicited pain. Furthermore, we do not know whether the five unilateral weight-bearing exercises described by Ayotte Stetts, Keenan, and Greenway (2007) could be performed in a pain-free manner despite EMG evidence illustrating robust activation of the gluteus maximus in contrast to low recruitment of the hamstrings.

The adductor magnus is classified as a primary extensor of the hip joint (Neumann, 2010). Because of the anatomical arrangement of the adductor magnus in the medial thigh, we chose not record surface EMG from this muscle during either supine exercise because placement of electrodes for the adductor muscles is not well defined (Lovell, Blanch, and Barnes, 2012) and technically challenging. We anticipated that the potential for cross talk between the hamstrings and adductor magnus would contaminate their respective EMG signals. Nevertheless, investigators recently (Lovell, Blanch, and Barnes, 2012) successfully examined activation levels of the four hip adductors including the adductor magnus using surface electrodes during six clinical examination tests. Because of its architecture, we believe that the level of muscle recruitment (% MVIC) from the adductor magnus would augment the hamstrings during the supine resisted hip extension exercise.

Conclusion

Magnitudes of muscle recruitment (% MVIC) were obtained from the primary hip extensors in healthy persons during performance of the unilateral supine bridge to neutral and supine resisted hip extension exercise. Hamstring activation was significantly reduced in the unilateral bridge compared to the supine hip extension exercise. The gluteus maximus to hamstring ratio was significantly greater in the unilateral bridge to neutral compared to the supine hip extension exercise reflecting preferential recruitment of the gluteus maximus. On the basis of surface EMG obtained from the present study, the supine resisted hip

extension exercise does not effectively maximize recruitment of the gluteus maximus and minimize recruitment of the hamstrings. Therefore, rehabilitation professionals should be cautious when selecting the supine resisted hip extension exercise as an appropriate method for strengthening the gluteus maximus in persons with anterior hip joint pain.

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Declaration of interest

The authors report no conflicts of interest.

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